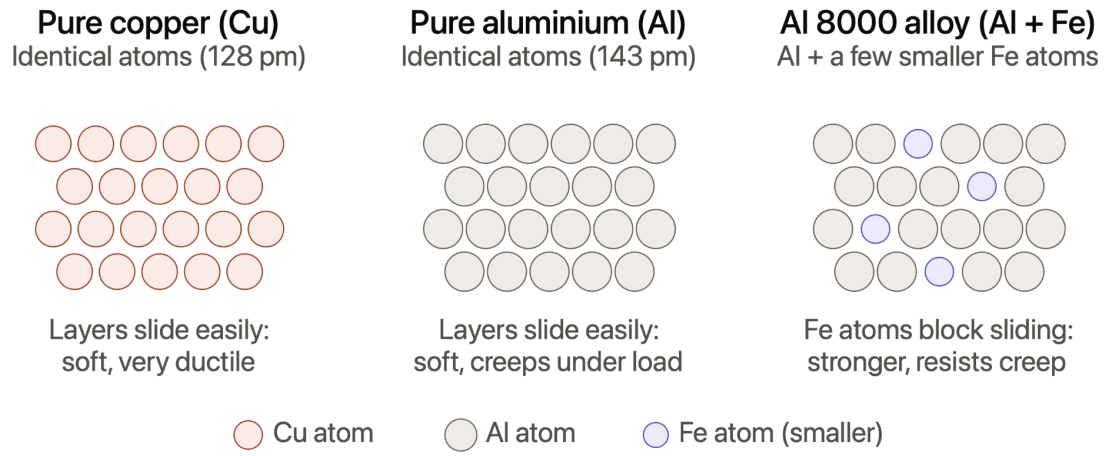
Design a better model

1. Intended application

A conductor for mains electrical cable, such as household wiring and appliance leads. It must carry current with little energy lost as heat, bend without breaking, and resist corrosion.

Materials compared

Property	Copper (pure)	Silver (pure)	Al 8000-series alloy (Al + about 0.4–1% Fe)
Electrical conductivity	about 58 MS/m	about 63 MS/m	about 35 MS/m (about 61% of copper)
Density (g/cm³)	8.96	10.49	~ 2.71
Tensile strength (MPa)	~ 220	about 125	~ 110–140
Ductility	Excellent	Excellent	Good
Creep resistance	Very good	Good	Good, much better than pure aluminium
Corrosion	Protective patina	Tarnishes	Oxide layer that insulates joints, so connectors need care
Cost	Moderate	Very high	Low



Model. Metallic bonding is the attraction between positive metal ions and a sea of delocalised electrons.

Pure metals (Cu, Ag, Al). The delocalised electrons are free to move, so the metal conducts. The bonding is non-directional, so layers of identical ions slide over each other without breaking the structure, and the metal can be drawn into wire. Cu and Ag have one delocalised electron per atom and conduct best. Al has three, but conducts less because the electrons scatter more as they move through the lattice.

Alloy (Al 8000 series). Iron atoms are a different size from aluminium atoms, so they distort the layers. The layers slide less easily, so the alloy is stronger and resists creep better. The distortion also scatters electrons slightly, so conductivity falls a little (about 61% of copper's). Conclusion. Pure copper suits small household wiring. The Al alloy suits large building cable, because it is about half the mass of copper for the same resistance. Silver conducts best, but costs too much for the small gain.

Recommendation

This depends on how the what application:

- **Small household wiring:** copper. It has the best balance of conductivity, ductility and reliable joints.
- **Large building or power-distribution cable:** the Al 8000-series alloy. Weight and cost matter most at that scale. The alloy gives about half the mass for the same resistance, and the alloying atoms fix pure aluminium's creep problem.
- **Silver:** rejected for ordinary cable, because the small conductivity gain doesn't justify the cost.